ml-face-blurring (/github/cgv1999/ml-face-blurring/tree/main)
/
HW5_MLOps_Чернигов_Григорий.ipynb (/github/cgv1999/ml-face-blurring/tree/main/HW5_MLOps_Чернигов_Григорий.ipyr

Домашнее задание 5. Обеспечение воспроизводимости эксперимента

In [1…

```
%%capture
!pip install feast psycopg psycopg_pool pandas scikit-learn -qqq
!pip install gdown tqdm -qqq
!pip install marimo -qqq
from tqdm import tqdm
import psycopg
import pandas as pd
import warnings
warnings.filterwarnings('ignore', category=DeprecationWarning)
!apt-get update -qq > /dev/null
!apt-get install -y yamllint > /dev/null
```

In [2…

```
from google.colab import userdata
import os
os.environ["tunatoken"] = userdata.get('tuna')
!curl -sSLf https://get.tuna.am | sh >/dev/null
!tuna config save-token $tunatoken
```

```
W: Skipping acquire of configured file 'main/source/Sources' as repository 'http
s://r2u.stat.illinois.edu/ubuntu jammy InRelease' does not seem to provide it (so
urces.list entry misspelt?)
W: Skipping acquire of configured file 'main/source/Sources' as repository 'http
s://r2u.stat.illinois.edu/ubuntu jammy InRelease' does not seem to provide it (so
urces.list entry misspelt?)
INFO[14:00:30] Token saved
```

1. Выяснить ограничения традиционных ноутбуков Collab при разработке ML-систем по сравнению с Marimo (https://github.com/marimo-team/marimo)

Ожидаемый артефакт: текст в текстовой ячейке

In [3…

```
print(f"Скопируйте адрес Public address, вставьте его в адресную строку браузера,
!sleep 5 && marimo tutorial ui --port 8010 --host 0.0.0.0 --headless --token-pass
```

```
Скопируйте адрес Public address, вставьте его в адресную строку браузера, добавьт
е ?access_token=password, нажмите Enter
INFO[14:00:52] Welcome to Tuna
INFO[14:00:54] Account: chernigov.gv@phystech.edu (Free)
INFO[14:00:54] Web Interface: http://127.0.0.1:4040

        Edit ui.py in your browser 📝

        ➜  URL: http://0.0.0.0:8010?access_token=password
        ➜  Network: http://172.28.0.12:8010?access_token=password

        💡 Tip: Generate thumbnails for a folder
                $ marimo export thumbnail folder/

ERRO[14:02:31] Timed out, retrying in 1s
ERRO[14:03:35] Timed out, retrying in 2s
INFO[14:03:39] Forwarding https://ljoswq-34-48-200-51.ru.tuna.am -> 127.0.0.1:801
0
INFO[14:03:39] Connection established
INFO[14:04:07] GET / — 303 See Other
INFO[14:04:07] GET / — 200 OK
INFO[14:04:08] GET /assets/gradient-yHQUC_QB.png — 200 OK
INFO[14:04:08] GET /assets/PTSans-Regular-Bam3NpBI.woff2 — 200 OK
INFO[14:04:08] GET /assets/tooltip-Gcwqb_SK.js — 200 OK
INFO[14:04:08] GET /assets/noise-60BoTA8O.png — 200 OK
INFO[14:04:08] GET /assets/Lora-VariableFont_wght-CZceb_kH.woff2 — 200 OK
INFO[14:04:08] GET /assets/FiraMono-Regular-CEsLFVD9.woff2 — 200 OK
INFO[14:04:08] GET /assets/FiraMono-Bold-C6PDArdf.woff2 — 200 OK
INFO[14:04:08] GET /assets/vega-loader.browser-DXARUlxo.js — 200 OK
INFO[14:04:08] GET /assets/FiraMono-Medium-D5MAsWEG.woff2 — 200 OK
INFO[14:04:08] GET /assets/useEventListener-BR0C1MaI.js — 200 OK
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INFO[14:04:08] GET /assets/check-BH35Ndha.js — 200 OK
INFO[14:04:08] GET /assets/precisionRound-CU2C3Vxx.js — 200 OK
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INFO[14:04:09] GET /assets/x-C-6liIBr.js — 200 OK
INFO[14:04:09] GET /assets/createLucideIcon-D5guW7EU.js — 200 OK
INFO[14:04:09] GET /assets/_baseFor-DKD1r8uL.js — 200 OK
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INFO[14:04:09] GET /assets/fullscreen-BDxedMYP.js — 200 OK
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INFO[14:04:09] GET /assets/clsx-Dz_KRrWq.js — 200 OK
INFO[14:04:09] GET /assets/defaultLocale-JieDVWC_.js — 200 OK
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INFO[14:04:09] GET /assets/select-5i7URBEn.js — 200 OK
INFO[14:04:09] GET /assets/badge-ChrPsdTW.js — 200 OK
INFO[14:04:09] GET /assets/dist-BoOh2kN5.js — 200 OK
INFO[14:04:09] GET /assets/button-D9nb17Rw.js — 200 OK
INFO[14:04:09] GET /assets/utils-Bq5kLQ87.js — 200 OK
INFO[14:04:09] GET /assets/invariant-BUdrueMv.js — 200 OK
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INFO[14:04:09] GET /assets/merge-_9WLTEyt.js — 200 OK
INFO[14:04:09] GET /assets/createReducer-1ePoj7v6.js — 200 OK
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INFO[14:04:21] GET /assets/velocity-D99sTvXl.js — 200 OK
INFO[14:04:21] GET /assets/textile-DhHo4rKl.js — 200 OK
INFO[14:04:21] GET /assets/vhdl-BAGmVXDV.js — 200 OK
INFO[14:04:21] GET /assets/ws-CtTwmdal.js — 200 OK
INFO[14:04:21] GET /assets/hooks-b3J1eFE-.js — 200 OK
INFO[14:04:21] GET /assets/xquery-B8Mguhly.js — 200 OK
INFO[14:04:21] GET /assets/sas-Be8P6y_s.js — 200 OK
INFO[14:04:21] GET /assets/sieve-CdflT_3W.js — 200 OK
INFO[14:04:21] GET /assets/factor-BbYo7zsD.js — 200 OK
INFO[14:04:21] GET /assets/idl-CjBvIccU.js — 200 OK
INFO[14:04:21] GET /assets/groovy-Ij7mXzwd.js — 200 OK
INFO[14:04:21] GET /assets/ttcn-CQzcRj0D.js — 200 OK
INFO[14:04:21] GET /assets/haxe-DYYwGmM8.js — 200 OK
INFO[14:04:21] GET /assets/runs-5odJFrfi.js — 200 OK
INFO[14:04:21] GET /assets/settings-DfFe0dWD.js — 200 OK
INFO[14:04:21] GET /assets/ttcn-cfg-jzTcqwRW.js — 200 OK
INFO[14:04:21] GET /assets/sparql-DEker8ZS.js — 200 OK
INFO[14:04:21] GET /assets/requests-BqcqF00P.js — 200 OK
INFO[14:04:21] GET /assets/turtle-BAqH2D2z.js — 200 OK
INFO[14:04:21] GET /assets/elm-D9Rly7Qq.js — 200 OK
INFO[14:04:21] GET /assets/haskell-BT_ZNgUJ.js — 200 OK
INFO[14:04:21] GET /assets/forth-C6jh4lHA.js — 200 OK
INFO[14:04:21] GET /assets/erlang-FZMRuv-7.js — 200 OK
INFO[14:04:21] GET /assets/verilog-XWBMMjwd.js — 200 OK
INFO[14:04:21] GET /assets/swift-D5F-FU3-.js — 200 OK
INFO[14:04:21] GET /assets/eiffel-BtJc6fqQ.js — 200 OK
INFO[14:04:21] GET /assets/z80-DhICfTcL.js — 200 OK
INFO[14:04:21] GET /assets/scheme-B9Zors0b.js — 200 OK
INFO[14:04:21] GET /assets/modelica-BsibU-7c.js — 200 OK
INFO[14:04:21] GET /assets/mumps-DIZYQE11.js — 200 OK
INFO[14:04:21] GET /assets/pascal-C9rPtth-.js — 200 OK
INFO[14:04:21] GET /assets/toml-CrP9HWVu.js — 200 OK
INFO[14:04:21] GET /assets/octave-BWtN_wao.js — 200 OK
INFO[14:04:21] GET /assets/stylus-DAJwnVOa.js — 200 OK
INFO[14:04:21] GET /assets/rpm-zbaZz_dk.js — 200 OK
INFO[14:04:21] GET /assets/mirc-6LiBC6Dw.js — 200 OK
INFO[14:04:21] GET /assets/link-CfEtM3Rl.js — 200 OK
INFO[14:04:21] GET /assets/r-DHkTNFL2.js — 200 OK
INFO[14:04:21] GET /assets/add-connection-dialog-HCShSlr3.js — 200 OK
INFO[14:04:21] GET /assets/lua-CT4vrRvN.js — 200 OK
INFO[14:04:21] GET /assets/ntriples-CwlUIE0o.js — 200 OK
INFO[14:04:21] GET /assets/circle-plus-BdiiIXu5.js — 200 OK
INFO[14:04:21] GET /assets/write-secret-modal-BfflSuL6.js — 200 OK
INFO[14:04:21] GET /assets/useCellActionButton-DYwx4pGE.js — 200 OK
INFO[14:04:21] GET /assets/perl-CF7pKLbl.js — 200 OK
INFO[14:04:21] GET /assets/nsis-Cr8jf68z.js — 200 OK
INFO[14:04:21] GET /assets/yacas-XW2NxGKw.js — 200 OK
INFO[14:04:21] GET /assets/pig-Bu0nn8JT.js — 200 OK
INFO[14:04:21] GET /assets/loro_wasm_bg-Ds40eH8K.js — 200 OK
INFO[14:04:21] GET /assets/github-raQvpeuZ.js — 200 OK
INFO[14:04:21] GET /assets/multi-icon-DQxqgfM2.js — 200 OK
INFO[14:04:21] GET /assets/layout-lxwMDUgI.js — 200 OK
INFO[14:04:21] GET /assets/mllike-orSu1SHG.js — 200 OK
INFO[14:04:21] GET /assets/mbox-DYDL20A1.js — 200 OK
INFO[14:04:21] GET /assets/esm-CbD72kxC.js — 200 OK
INFO[14:04:21] GET /assets/livescript-D7jeebX7.js — 200 OK
INFO[14:04:21] GET /assets/useNotebookActions-Dga3qEHF.js — 200 OK
INFO[14:04:21] GET /assets/julia-U0NSUEwm.js — 200 OK
INFO[14:04:21] GET /assets/mathematica-DmzlinOc.js — 200 OK
INFO[14:04:21] GET /assets/oz-CJMVKlqL.js — 200 OK
INFO[14:04:21] GET /assets/save-Clg5dMoP.js — 200 OK
INFO[14:04:21] GET /assets/protobuf-Vz3fs9Hb.js — 200 OK
INFO[14:04:21] GET /assets/powershell-Bzwpi8yQ.js — 200 OK
INFO[14:04:21] GET /assets/javascript-DQz7szA1.js — 200 OK
```

```
INFO[14:04:21] GET /assets/mscgen-vX4NsSXf.js — 200 OK
INFO[14:04:21] GET /assets/properties-Cfh_Lj0j.js — 200 OK
INFO[14:04:21] GET /assets/links-BHtUMVOq.js — 200 OK
INFO[14:04:21] GET /assets/dist-HW7R98td.js — 200 OK
INFO[14:04:21] GET /assets/apl-T0uH_RjM.js — 200 OK
INFO[14:04:21] GET /assets/dist-BtG3xAGt.js — 200 OK
INFO[14:04:21] GET /assets/cobol-BWxHqof4.js — 200 OK
INFO[14:04:21] GET /assets/coffeescript-nKvg8y_U.js — 200 OK
INFO[14:04:21] GET /assets/web-vitals-BaJPM15X.js — 200 OK
INFO[14:04:21] GET /assets/dist-C46P-i95.js — 200 OK
INFO[14:04:21] GET /assets/dist-D6wwIjcW.js — 200 OK
INFO[14:04:21] GET /assets/commonlisp-Dm562cBV.js — 200 OK
INFO[14:04:21] GET /assets/clojure-Df1U5nhC.js — 200 OK
INFO[14:04:21] GET /assets/youtube-3lRHw8NU.js — 200 OK
INFO[14:04:21] GET /assets/clike-BK27e5qn.js — 200 OK
INFO[14:04:21] GET /assets/circle-check-C17zXg7L.js — 200 OK
INFO[14:04:21] GET /assets/crystal-CrCmK2Cb.js — 200 OK
INFO[14:04:21] GET /assets/cell-editor-BMjfXh0J.js — 200 OK
INFO[14:04:21] GET /assets/workflow-DYcZF-oV.js — 200 OK
INFO[14:04:21] GET /assets/dist-0xasIeT1.js — 200 OK
INFO[14:04:21] GET /assets/cmake-DrCf0yvp.js — 200 OK
INFO[14:04:21] GET /assets/dist-ooLAQ9IK.js — 200 OK
INFO[14:04:21] GET /assets/CellStatus-BTL_sgUY.js — 200 OK
INFO[14:04:21] GET /assets/asciiarmor-D3ol7c2G.js — 200 OK
INFO[14:04:21] GET /assets/brainfuck-exn1QR7H.js — 200 OK
INFO[14:04:21] GET /assets/marimo-icons-BSB-Z_W3.js — 200 OK
INFO[14:04:21] GET /assets/dist-OWcDq-O3.js — 200 OK
INFO[14:04:21] GET /assets/add-cell-with-ai-qEfAzC8R.js — 200 OK
INFO[14:04:21] GET /assets/asn1-DA-fXXsk.js — 200 OK
INFO[14:04:21] GET /assets/dist-DmZscTVZ.js — 200 OK
INFO[14:04:21] GET /assets/dist-CsEqrZxx.js — 200 OK
INFO[14:04:21] GET /assets/ai-model-dropdown-CYx9ZfdS.js — 200 OK
INFO[14:04:21] GET /assets/edit-page-Bxa5DWoE.js — 200 OK
INFO[14:04:22] GET /public-files-sw.js — 200 OK
INFO[14:04:22] GET /assets/loro_wasm_bg-WKwciKJa.wasm — 200 OK
INFO[14:04:24] GET /health — 200 OK
INFO[14:04:24] GET /assets/command-palette-DSSR50KF.js — 200 OK
INFO[14:04:27] POST /api/kernel/instantiate — 200 OK
INFO[14:04:27] GET /health — 200 OK
INFO[14:04:27] GET /api/lsp/health — 200 OK
INFO[14:04:27] GET /api/usage — 200 OK
INFO[14:04:30] POST /api/document/transaction — 200 OK
INFO[14:04:34] GET /api/usage — 200 OK
INFO[14:04:44] GET /api/usage — 200 OK
INFO[14:04:54] GET /api/usage — 200 OK
INFO[14:05:16] GET /ws — 101 Switching Protocols
INFO[14:05:16] Bye

        Thanks for using marimo! 🌊🍃

^C
```

bsdkgx-34-48-200-51.ru.tuna.am ui.py

UI Elements

One of marimo's most powerful features is its first-class support for interactive user interface (UI) elements: interacting with a UI element will automatically run cells that reference it.

```
slider = mo.ui.slider(start=1, step=10, step=1)
slider

mo.md(
    f"""
    The `marimo.ui` module has a library of pre-built elements.

    For example, here's a `slider`: {slider}
    """
)
```

The `marimo.ui` module has a library of pre-built elements.

For example, here's a `slider`:

```
# Ячейка 2 - зависит от слайдера
value = slider.value
mo.md(f"## Текущее значение слайдера: **{value}**")
```

Текущее значение слайдера: 1

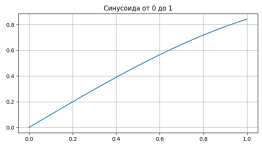
```
# Ячейка 3 - тоже зависит от слайдера
squared = slider.value ** 2
mo.md(f"### Квадрат значения: **{squared}**")
```

Квадрат значения: 1

```
# Ячейка 4 - график, зависящий от слайдера
import matplotlib.pyplot as plt
import numpy as np

x = np.linspace(0, slider.value, 100)
y = np.sin(x)

plt.figure(figsize=(8, 4))
plt.plot(x, y)
plt.title(f"Синусоида от 0 до {slider.value}")
plt.grid(True)
plt.gca()
```



marimo.ui

bsdkgx-34-48-200-51.ru.tuna.am ui.py

UI Elements

One of marimo's most powerful features is its first-class support for interactive user interface (UI) elements: interacting with a UI element will automatically run cells that reference it.

```
slider = mo.ui.slider(start=1, step=10, step=1)
slider

mo.md(
    f"""
    The `marimo.ui` module has a library of pre-built elements.

    For example, here's a `slider`: {slider}
    """
)
```

The `marimo.ui` module has a library of pre-built elements.

For example, here's a `slider`:

```
# Ячейка 2 - зависит от слайдера
value = slider.value
mo.md(f"## Текущее значение слайдера: **{value}**")
```

Текущее значение слайдера: 4

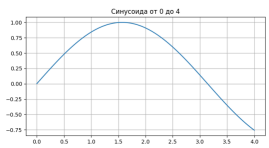
```
# Ячейка 3 - тоже зависит от слайдера
squared = slider.value ** 2
mo.md(f"### Квадрат значения: **{squared}**")
```

Квадрат значения: 16

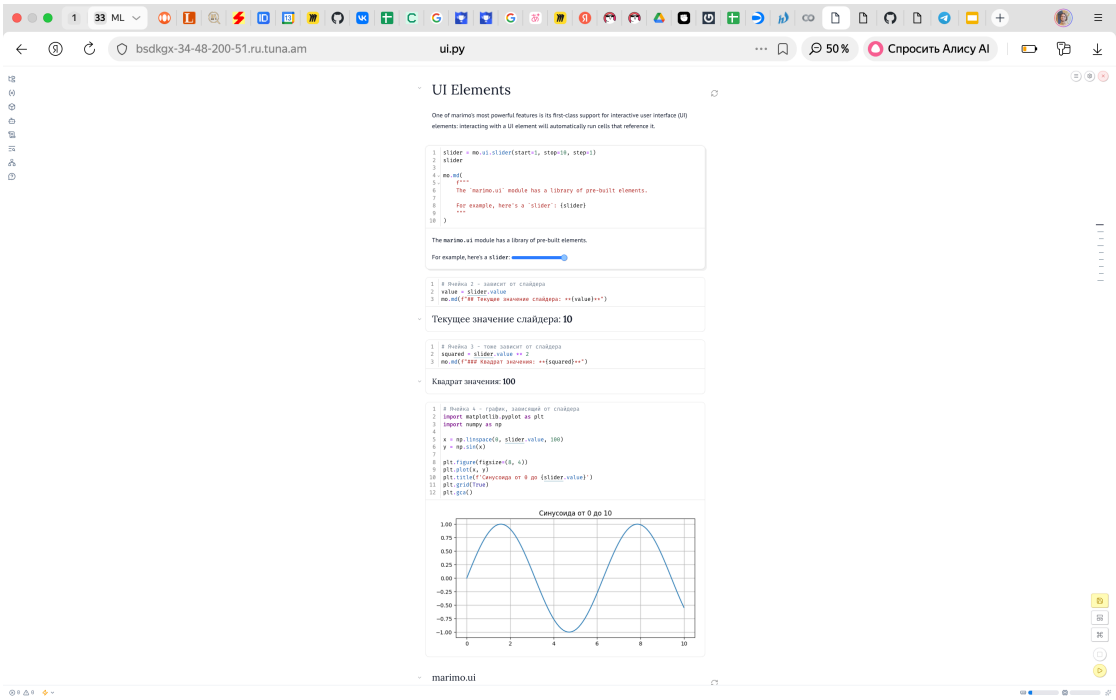
```
# Ячейка 4 - график, зависящий от слайдера
import matplotlib.pyplot as plt
import numpy as np

x = np.linspace(0, slider.value, 100)
y = np.sin(x)

plt.figure(figsize=(8, 4))
plt.plot(x, y)
plt.title(f"Синусоида от 0 до {slider.value}")
plt.grid(True)
plt.gca()
```



marimo.ui



Вау. Это действительно здорово - при изменении значения UI-элемента (слайдера) все зависимые ячейки автоматически пересчитываются без явного перезапуска.

Нам не нужно перезапускать ядро, нам не нужно менять данные в ячейке - у нас есть согласованность данных, их связность с другими ячейками - при смене переменной в одной ячейке все автоматически меняется!

Это и есть реактивные вычисления - подход, знакомый по Excel, но реализованный для полноценного программирования на Python.

(UI работает только с включенным ВПН...)

2. Обеспечить воспроизводимость эксперимента, используя DVC/MLflow

- Инициализируйте Git и DVC.
- Добавьте небольшой датасет (например, Iris, Titanic или Wine).
- **Не храните данные в Git, используйте DVC**.
- Создайте пайплайн DVC минимум в две стадии.

Ожидаемый артефакт: bash-скрипт в ячейке

In [4…

```
import pandas as pd
df = pd.read_csv("https://gist.githubusercontent.com/netj/8836201/raw/6f9306ad213
df.dropna(inplace=True)
df.to_csv("energydata_complete_v1.csv")
```

In [5…

```
!pip install dvc -q
```

In [6…

```
%%bash
# 1. Инициализация Git
git init && git checkout -b master
git config --global user.email "publicemail@example.com"
git config --global user.name "publicname"

# 2. Инициализация DVC
dvc config core.analytics false
dvc init -f
dvc remote add -d myremote datasets -f

# 3. Добавление данных в DVC
dvc add energydata_complete_v1.csv && dvc commit
git add energydata_complete_v1.csv.dvc .dvc/config
git commit -m "initial_data"

# 4. Пропускаем push (всё равно не сработает без настроенного удаленного хранилища

# 5. Создаем скрипты для пайплайна
cat > script1.py << 'EOF'
import pandas as pd
df = pd.read_csv("energydata_complete_v1.csv")
# Предобработка данных
df.to_csv("processed_data.csv", index=False)
print("Preprocessing completed")
EOF

cat > script2.py << 'EOF'
import pandas as pd
df = pd.read_csv("processed_data.csv")
# Постобработка
df.to_csv("final_output.csv", index=False)
print("Postprocessing completed")
EOF

# 6. Очистка экспериментов
dvc exp remove -A
dvc queue remove --all

# 7. Создание пайплайна
dvc stage add --name=preprocessing \
    --deps energydata_complete_v1.csv \
    --outs processed_data.csv \
    --force \
    python script1.py

dvc stage add --name=postprocessing \
    --deps processed_data.csv \
    --outs final_output.csv \
    --force \
    python script2.py

# 8. Запуск пайплайна
dvc repro

# 9. Сохранение эксперимента
git checkout -b dvc-241106
dvc exp save --name first -m "241106"
dvc exp show
```

```
Reinitialized existing Git repository in /content/.git/
Initialized DVC repository.

You can now commit the changes to git.

+---------------------------------------------------------------------+
|                                                                     |
|        DVC has enabled anonymous aggregate usage analytics.         |
|     Read the analytics documentation (and how to opt-out) here:     |
|             <https://dvc.org/doc/user-guide/analytics>              |
|                                                                     |
+---------------------------------------------------------------------+

What's next?
------------
- Check out the documentation: <https://dvc.org/doc>
- Get help and share ideas: <https://dvc.org/chat>
- Star us on GitHub: <https://github.com/treeverse/dvc>
Setting 'myremote' as a default remote.

To track the changes with git, run:

        git add energydata_complete_v1.csv.dvc

To enable auto staging, run:

        dvc config core.autostage true
On branch dvc-241106
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .config/
        local_repo/
        mlruns/
        sample_data/

nothing added to commit but untracked files present (use "git add" to track)
Removed experiments: 'first'
No tasks found named []
Added stage 'preprocessing' in 'dvc.yaml'

To track the changes with git, run:

        git add dvc.yaml

To enable auto staging, run:

        dvc config core.autostage true
Added stage 'postprocessing' in 'dvc.yaml'

To track the changes with git, run:

        git add dvc.yaml

To enable auto staging, run:

        dvc config core.autostage true
'energydata_complete_v1.csv.dvc' didn't change, skipping
Stage 'preprocessing' didn't change, skipping
Stage 'postprocessing' didn't change, skipping
Data and pipelines are up to date.
Experiment has been saved as: first
 ──────────────
```

```
  workspace             -          9d69e129ab72daf11e3712a9d8895eb5   926993d0c68
8ab1fc0d5a1aac47ee594
  dvc-241106            12:10 PM   9d69e129ab72daf11e3712a9d8895eb5   926993d0c68
8ab1fc0d5a1aac47ee594
  └── 2aadbeb [first]   02:12 PM   9d69e129ab72daf11e3712a9d8895eb5   926993d0c68
8ab1fc0d5a1aac47ee594
```

```
fatal: A branch named 'master' already exists.
fatal: A branch named 'dvc-241106' already exists.
WARNING: The following untracked files were present in the workspace before savin
g but will not be included in the experiment commit:
        .config/.last_survey_prompt.yaml, .config/.last_update_check.json, .confi
g/default_configs.db, .config/hidden_gcloud_config_universe_descriptor_data_cache
_configs.db, .config/.last_opt_in_prompt.yaml, .config/active_config, .config/con
fig_sentinel, .config/gce, .config/logs/2026.04.16/13.27.28.140888.log, .config/l
ogs/2026.04.16/13.28.06.952902.log, .config/logs/2026.04.16/13.28.22.950479.log,
.config/logs/2026.04.16/13.27.53.684484.log, .config/logs/2026.04.16/13.28.21.827
521.log, .config/logs/2026.04.16/13.28.05.189231.log, .config/configurations/conf
ig_default, local_repo/__init__.py, local_repo/README.md, local_repo/.gitignore,
local_repo/feature_repo/__init__.py, local_repo/feature_repo/test_workflow.py, lo
cal_repo/feature_repo/feature_store.yaml, local_repo/feature_repo/feature_definit
ions.py, local_repo/feature_repo/data/driver_stats.parquet, local_repo/feature_re
po/data/registry.db, mlruns/models/это название модели/meta.yaml, mlruns/models/э
то название модели/version-3/meta.yaml, mlruns/models/это название модели/version
-1/meta.yaml, mlruns/models/это название модели/version-2/meta.yaml, mlruns/30177
7073482157687/meta.yaml, mlruns/301777073482157687/19eca70a1e4c41faa50e148b6712c0
72/meta.yaml, mlruns/301777073482157687/19eca70a1e4c41faa50e148b6712c072/artifact
s/estimator.html, mlruns/301777073482157687/19eca70a1e4c41faa50e148b6712c072/arti
facts/metric_info.json, mlruns/301777073482157687/19eca70a1e4c41faa50e148b6712c07
2/params/bootstrap, mlruns/301777073482157687/19eca70a1e4c41faa50e148b6712c072/pa
rams/min_weight_fraction_leaf, mlruns/301777073482157687/19eca70a1e4c41faa50e148b
6712c072/params/n_jobs, mlruns/301777073482157687/19eca70a1e4c41faa50e148b6712c07
2/params/n_estimators, mlruns/301777073482157687/19eca70a1e4c41faa50e148b6712c07
2/params/max_features, mlruns/301777073482157687/19eca70a1e4c41faa50e148b6712c07
2/params/min_samples_split, mlruns/301777073482157687/19eca70a1e4c41faa50e148b671
2c072/params/min_samples_leaf, mlruns/301777073482157687/19eca70a1e4c41faa50e148b
6712c072/params/min_impurity_decrease, mlruns/301777073482157687/19eca70a1e4c41fa
a50e148b6712c072/params/oob_score, mlruns/301777073482157687/19eca70a1e4c41faa50e
148b6712c072/params/ccp_alpha, mlruns/301777073482157687/19eca70a1e4c41faa50e148b
6712c072/params/verbose, mlruns/301777073482157687/19eca70a1e4c41faa50e148b6712c0
72/params/random_state, mlruns/301777073482157687/19eca70a1e4c41faa50e148b6712c07
2/params/max_leaf_nodes, mlruns/301777073482157687/19eca70a1e4c41faa50e148b6712c0
72/params/monotonic_cst, mlruns/301777073482157687/19eca70a1e4c41faa50e148b6712c0
72/params/max_samples, mlruns/301777073482157687/19eca70a1e4c41faa50e148b6712c07
2/params/warm_start, mlruns/301777073482157687/19eca70a1e4c41faa50e148b6712c072/p
arams/max_depth, mlruns/301777073482157687/19eca70a1e4c41faa50e148b6712c072/param
s/criterion, mlruns/301777073482157687/19eca70a1e4c41faa50e148b6712c072/inputs/93
4788434dcce8805abe3671504238f7/meta.yaml, mlruns/301777073482157687/19eca70a1e4c4
1faa50e148b6712c072/inputs/59af08b00936821c389001d3ee2cd98d/meta.yaml, mlruns/301
777073482157687/19eca70a1e4c41faa50e148b6712c072/inputs/2aabfad4733d84ee768fa090d
137540b/meta.yaml, mlruns/301777073482157687/19eca70a1e4c41faa50e148b6712c072/tag
s/mlflow.user, mlruns/301777073482157687/19eca70a1e4c41faa50e148b6712c072/tags/ml
flow.runName, mlruns/301777073482157687/19eca70a1e4c41faa50e148b6712c072/tags/mlf
low.source.name, mlruns/301777073482157687/19eca70a1e4c41faa50e148b6712c072/tags/
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sv
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In [7…

```
%%bash
# Возвращаемся на ветку с пайплайном
git checkout dvc-241106

# Добавляем ВСЕ файлы пайплайна в Git
git add dvc.yaml dvc.lock script1.py script2.py .gitignore
git commit -m "Complete DVC pipeline"

# Проверяем статус
git status
dvc exp show
```

```
On branch dvc-241106
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .config/
        local_repo/
        mlruns/
        sample_data/

nothing added to commit but untracked files present (use "git add" to track)
On branch dvc-241106
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .config/
        local_repo/
        mlruns/
        sample_data/

nothing added to commit but untracked files present (use "git add" to track)
 ──────────────────────────────────────────────────────────────────────────────
──────────────────────
  Experiment            Created    energydata_complete_v1.csv         processed_d
ata.csv
 ──────────────────────────────────────────────────────────────────────────────
──────────────────────
  workspace             -          9d69e129ab72daf11e3712a9d8895eb5   926993d0c68
8ab1fc0d5a1aac47ee594
  dvc-241106            12:10 PM   9d69e129ab72daf11e3712a9d8895eb5   926993d0c68
8ab1fc0d5a1aac47ee594
  └── 2aadbeb [first]   02:12 PM   9d69e129ab72daf11e3712a9d8895eb5   926993d0c68
8ab1fc0d5a1aac47ee594
 ──────────────────────────────────────────────────────────────────────────────
──────────────────────
Already on 'dvc-241106'
```

Теперь проверим воспроизводимость пайплайна:

In [8…

```
%%bash
rm -f processed_data.csv final_output.csv

echo "Файлы удалены. Запускаем dvc repro"
dvc repro

echo ""
echo "Проверяем, что файлы восстановлены"
ls -la processed_data.csv final_output.csv
```

```
Файлы удалены. Запускаем dvc repro
'energydata_complete_v1.csv.dvc' didn't change, skipping
Stage 'preprocessing' is cached - skipping run, checking out outputs

Stage 'postprocessing' is cached - skipping run, checking out outputs
Use `dvc push` to send your updates to remote storage.

Проверяем, что файлы восстановлены
-rw-r--r-- 1 root root 4359 Apr 26 14:12 final_output.csv
-rw-r--r-- 1 root root 4359 Apr 26 14:12 processed_data.csv
```

Все отлично, воспроизводимость доказана.

Команда dvc repro - и всё воспроизводится автоматически.

Данные версионированы, этапы зафиксированы.

In [9…

```
!pip install mlflow scikit-learn -q
```

In [1…

```
!sleep 10
import os
import mlflow
import pandas as pd

os.environ['MLFLOW_S3_ENDPOINT_URL'] = 'http://147.45.147.94:9000'
os.environ['AWS_ACCESS_KEY_ID'] = 'mlflow'
os.environ['AWS_SECRET_ACCESS_KEY'] = 'password'
os.environ['MLFLOW_TRACKING_URI'] = 'http://147.45.147.94:5000'
import mlflow
import mlflow.sklearn
from mlflow.models import infer_signature

# локальный режим вместо недоступного сервера
mlflow.set_tracking_uri("file:///content/mlruns")
mlflow.set_experiment("KPI tracing")
mlflow.autolog()

from sklearn.datasets import make_regression
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_squared_error
from sklearn.model_selection import train_test_split

df = pd.read_csv("energydata_complete_v1.csv")
dataset_source_url = "energydata_complete_v1.csv"

@mlflow.trace(name = "отображение трассировки", attributes = { "key": "value" })
def demonstrate_trace(inp=1): return 1

with mlflow.start_run(run_name="mlflow.start_run") as run:
    X, y = make_regression(n_features=4, n_informative=2, random_state=0, shuffle=
    X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size=0.2, random_state=42
    )
    params = {"max_depth": 10, "random_state": 42}

    model = RandomForestRegressor(**params)
    model.fit(X_train, y_train)

    y_pred = model.predict(X_test)
    signature = infer_signature(X_test, y_pred)

    mlflow.log_params(params)
    mlflow.log_metrics({"mse": mean_squared_error(y_test, y_pred)})

    dataset = mlflow.data.from_pandas(df,
                                      source=dataset_source_url,
                                      name="это набор данных",
                                      targets='variety')  # Исправлено: колонка из 1
    mlflow.log_input(dataset, context="training")

    mlflow.sklearn.log_model(
        sk_model=model,
        artifact_path="sklearn-model",
        signature=signature,
        registered_model_name="это название модели",
    )

demonstrate_trace()
print("Готово!")
```

```
/usr/local/lib/python3.12/dist-packages/mlflow/tracking/_tracking_service/utils.p
y:184: FutureWarning: The filesystem tracking backend (e.g., './mlruns') is depre
cated as of February 2026. Consider transitioning to a database backend (e.g., 's
qlite:///mlflow.db') to take advantage of the latest MLflow features. See http
s://mlflow.org/docs/latest/self-hosting/migrate-from-file-store for migration gui
dance.
  return FileStore(store_uri, store_uri)
2026/04/26 14:13:32 WARNING mlflow.spark: With Pyspark >= 3.2, PYSPARK_PIN_THREAD
environment variable must be set to false for Spark datasource autologging to wor
k.
2026/04/26 14:13:32 INFO mlflow.tracking.fluent: Autologging successfully enabled
for pyspark.
2026/04/26 14:13:33 INFO mlflow.tracking.fluent: Autologging successfully enabled
for sklearn.
2026/04/26 14:13:33 WARNING mlflow.sklearn: Saving scikit-learn models in the pic
kle or cloudpickle format requires exercising caution because these formats rely
on Python's object serialization mechanism, which can execute arbitrary code duri
ng deserialization. The recommended safe alternative is the 'skops' format. For m
ore information, see: https://scikit-learn.org/stable/model_persistence.html
/usr/local/lib/python3.12/dist-packages/mlflow/data/dataset_source_registry.py:14
8: UserWarning: The specified dataset source can be interpreted in multiple ways:
LocalArtifactDatasetSource, LocalArtifactDatasetSource. MLflow will assume that t
his is a LocalArtifactDatasetSource source.
  return _dataset_source_registry.resolve(
/usr/local/lib/python3.12/dist-packages/mlflow/types/utils.py:440: UserWarning: H
int: Inferred schema contains integer column(s). Integer columns in Python cannot
represent missing values. If your input data contains missing values at inference
time, it will be encoded as floats and will cause a schema enforcement error. The
best way to avoid this problem is to infer the model schema based on a realistic
data sample (training dataset) that includes missing values. Alternatively, you c
an declare integer columns as doubles (float64) whenever these columns may have m
issing values. See `Handling Integers With Missing Values <https://www.mlflow.or
g/docs/latest/models.html#handling-integers-with-missing-values>`_ for more detai
ls.
  warnings.warn(
2026/04/26 14:13:39 WARNING mlflow.models.model: `artifact_path` is deprecated. P
lease use `name` instead.
2026/04/26 14:13:39 WARNING mlflow.sklearn: Saving scikit-learn models in the pic
kle or cloudpickle format requires exercising caution because these formats rely
on Python's object serialization mechanism, which can execute arbitrary code duri
ng deserialization. The recommended safe alternative is the 'skops' format. For m
ore information, see: https://scikit-learn.org/stable/model_persistence.html
Готово!
/usr/local/lib/python3.12/dist-packages/mlflow/tracking/_model_registry/utils.py:
220: FutureWarning: The filesystem model registry backend (e.g., './mlruns') is d
eprecated as of February 2026. Consider transitioning to a database backend (e.
g., 'sqlite:///mlflow.db') to take advantage of the latest MLflow features. See h
ttps://mlflow.org/docs/latest/self-hosting/migrate-from-file-store for migration
guidance.
  return FileStore(store_uri)
Registered model 'это название модели' already exists. Creating a new version of
this model...
Created version '4' of model 'это название модели'.
```

Выводы:

Мы создали некий журнал (MLFlow), с помощью которого контролируем обучение модели - можем явно сказать, на каком датасете, когда именно, какая версия модели была использована.

Так как сервер не работает - складываем все локально в /content/mlruns.

В реальной жизни был бы Tracking Server для метаданных и S3-хранилище для файлов моделей. Коллега заходит на сервер, видит пятничную модель, смотрит метрики и скачивает ее из S3.

3. Создать хранилище признаков (Feature Store)

Используйте материалы семинара (https://colab.research.google.com/drive/1TNd3m8-rPrTx1OjRmCnOxUGFueLbXyfe?usp=sharing), чтобы создать хранилище признаков на основе шаблона `--template [local или postgres]`

Нужно сконфигурировать хранилище признаков в yaml-файле.

Ожидаемый артефакт: yaml-файл в ячейке

In [1…

```
!feast init --help
!feast init local_repo --template local
```

```
/usr/local/lib/python3.12/dist-packages/matplotlib/_fontconfig_pattern.py:64: Pyp
arsingDeprecationWarning: 'oneOf' deprecated - use 'one_of'
  prop = Group((name + Suppress("=") + comma_separated(value)) | oneOf(_CONSTANT
S))
/usr/local/lib/python3.12/dist-packages/matplotlib/_fontconfig_pattern.py:85: Pyp
arsingDeprecationWarning: 'parseString' deprecated - use 'parse_string'
  parse = parser.parseString(pattern)
/usr/local/lib/python3.12/dist-packages/matplotlib/_fontconfig_pattern.py:89: Pyp
arsingDeprecationWarning: 'resetCache' deprecated - use 'reset_cache'
  parser.resetCache()
/usr/local/lib/python3.12/dist-packages/matplotlib/_mathtext.py:45: PyparsingDepr
ecationWarning: 'enablePackrat' deprecated - use 'enable_packrat'
  ParserElement.enablePackrat()
In /usr/local/lib/python3.12/dist-packages/matplotlib/mpl-data/stylelib/classic.m
plstyle: 'parseString' deprecated - use 'parse_string'
In /usr/local/lib/python3.12/dist-packages/matplotlib/mpl-data/stylelib/classic.m
plstyle: 'resetCache' deprecated - use 'reset_cache'
Usage: feast init [OPTIONS] [PROJECT_DIRECTORY]

  Create a new Feast repository

Options:
  -m, --minimal                   Create an empty project repository
  -t, --template [local|gcp|aws|snowflake|spark|postgres|hbase|cassandra|hazelcas
t|couchbase|milvus|ray|ray_rag|pytorch_nlp]
                                  Specify a template for the created project
  --repo-path TEXT                Directory path where the repository will be
                                  created (default: create subdirectory with
                                  project name)
  --help                          Show this message and exit.
/usr/local/lib/python3.12/dist-packages/matplotlib/_fontconfig_pattern.py:64: Pyp
arsingDeprecationWarning: 'oneOf' deprecated - use 'one_of'
  prop = Group((name + Suppress("=") + comma_separated(value)) | oneOf(_CONSTANT
S))
/usr/local/lib/python3.12/dist-packages/matplotlib/_fontconfig_pattern.py:85: Pyp
arsingDeprecationWarning: 'parseString' deprecated - use 'parse_string'
  parse = parser.parseString(pattern)
/usr/local/lib/python3.12/dist-packages/matplotlib/_fontconfig_pattern.py:89: Pyp
arsingDeprecationWarning: 'resetCache' deprecated - use 'reset_cache'
  parser.resetCache()
/usr/local/lib/python3.12/dist-packages/matplotlib/_mathtext.py:45: PyparsingDepr
ecationWarning: 'enablePackrat' deprecated - use 'enable_packrat'
  ParserElement.enablePackrat()
In /usr/local/lib/python3.12/dist-packages/matplotlib/mpl-data/stylelib/classic.m
plstyle: 'parseString' deprecated - use 'parse_string'
In /usr/local/lib/python3.12/dist-packages/matplotlib/mpl-data/stylelib/classic.m
plstyle: 'resetCache' deprecated - use 'reset_cache'

Creating a new Feast repository in /content/local_repo.
```

In [1…

```
%%writefile /content/local_repo/feature_repo/feature_store.yaml
---
project: local_repo
registry: data/registry.db
provider: local
online_store:
    type: sqlite
    path: data/online_store.db
entity_key_serialization_version: 3
auth:
    type: no_auth
```

```
Overwriting /content/local_repo/feature_repo/feature_store.yaml
```

In [1…

```
!cat /content/local_repo/feature_repo/feature_store.yaml
```

```
---
project: local_repo
registry: data/registry.db
provider: local
online_store:
    type: sqlite
    path: data/online_store.db
entity_key_serialization_version: 3
auth:
    type: no_auth
```

Артефакт создан, задание выполнено.

In [1…

```
%%bash
yamllint /content/local_repo/feature_repo/feature_store.yaml
```

In [1…

```
%%bash
yamllint /content/local_repo/feature_repo/feature_store.yaml
```

In [1…

```
!cd /content/local_repo/feature_repo && feast ui --host 0.0.0.0 --port 8889 & sle
```

```
INFO[14:18:23] Welcome to Tuna
INFO[14:18:24] Account: chernigov.gv@phystech.edu (Free)
INFO[14:18:24] Web Interface: http://127.0.0.1:4040
/usr/local/lib/python3.12/dist-packages/matplotlib/_fontconfig_pattern.py:64: Pyp
arsingDeprecationWarning: 'oneOf' deprecated - use 'one_of'
  prop = Group((name + Suppress("=") + comma_separated(value)) | oneOf(_CONSTANT
S))
/usr/local/lib/python3.12/dist-packages/matplotlib/_fontconfig_pattern.py:85: Pyp
arsingDeprecationWarning: 'parseString' deprecated - use 'parse_string'
  parse = parser.parseString(pattern)
/usr/local/lib/python3.12/dist-packages/matplotlib/_fontconfig_pattern.py:89: Pyp
arsingDeprecationWarning: 'resetCache' deprecated - use 'reset_cache'
  parser.resetCache()
/usr/local/lib/python3.12/dist-packages/matplotlib/_mathtext.py:45: PyparsingDepr
ecationWarning: 'enablePackrat' deprecated - use 'enable_packrat'
  ParserElement.enablePackrat()
In /usr/local/lib/python3.12/dist-packages/matplotlib/mpl-data/stylelib/classic.m
plstyle: 'parseString' deprecated - use 'parse_string'
In /usr/local/lib/python3.12/dist-packages/matplotlib/mpl-data/stylelib/classic.m
plstyle: 'resetCache' deprecated - use 'reset_cache'
/usr/local/lib/python3.12/dist-packages/feast/ui_server.py:45: DeprecationWarnin
g:
        on_event is deprecated, use lifespan event handlers instead.

        Read more about it in the
        [FastAPI docs for Lifespan Events](https://fastapi.tiangolo.com/advanced/
events/).

  @app.on_event("shutdown")
/usr/local/lib/python3.12/dist-packages/fastapi/applications.py:4656: Deprecation
Warning:
        on_event is deprecated, use lifespan event handlers instead.

        Read more about it in the
        [FastAPI docs for Lifespan Events](https://fastapi.tiangolo.com/advanced/
events/).

  return self.router.on_event(event_type)  # ty: ignore[deprecated]
/usr/local/lib/python3.12/dist-packages/websockets/legacy/__init__.py:6: Deprecat
ionWarning: websockets.legacy is deprecated; see https://websockets.readthedocs.i
o/en/stable/howto/upgrade.html for upgrade instructions
  warnings.warn(  # deprecated in 14.0 - 2024-11-09
/usr/local/lib/python3.12/dist-packages/uvicorn/protocols/websockets/websockets_i
mpl.py:17: DeprecationWarning: websockets.server.WebSocketServerProtocol is depre
cated
  from websockets.server import WebSocketServerProtocol
INFO:     Started server process [61130]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
INFO:     Uvicorn running on http://0.0.0.0:8889 (Press CTRL+C to quit)
ERRO[14:19:32] Timed out, retrying in 1s
ERRO[14:20:35] Timed out, retrying in 2s
WARN[14:21:08] Connection lost, reconnecting: 127.0.0.1:40371
INFO[14:21:08] Reconnect cancelled: 127.0.0.1:40371
ERRO[14:21:08] Timed out, retrying in 4s
INFO:     Shutting down
INFO:     Waiting for application shutdown.
INFO:     Application shutdown complete.
INFO:     Finished server process [61130]
ERRO[14:22:10] Timed out, retrying in 8s
INFO[14:22:10] Bye
```

Не работает из-за санкций.

Решение - создадим порт вручную из лк Tuna, выберем Нидерланды и порт 8889.

In [1…

```
!cd /content/local_repo/feature_repo && feast ui --host 0.0.0.0 --port 8889 & sle
```

```
INFO[14:29:27] Welcome to Tuna
INFO[14:29:28] Account: chernigov.gv@phystech.edu (Free)
INFO[14:29:28] Web Interface: http://127.0.0.1:4040
INFO[14:29:29] Forwarding https://43t2p8-34-48-200-51.nl.tuna.am -> 127.0.0.1:888
9
/usr/local/lib/python3.12/dist-packages/matplotlib/_fontconfig_pattern.py:64: Pyp
arsingDeprecationWarning: 'oneOf' deprecated - use 'one_of'
  prop = Group((name + Suppress("=") + comma_separated(value)) | oneOf(_CONSTANT
S))
/usr/local/lib/python3.12/dist-packages/matplotlib/_fontconfig_pattern.py:85: Pyp
arsingDeprecationWarning: 'parseString' deprecated - use 'parse_string'
  parse = parser.parseString(pattern)
/usr/local/lib/python3.12/dist-packages/matplotlib/_fontconfig_pattern.py:89: Pyp
arsingDeprecationWarning: 'resetCache' deprecated - use 'reset_cache'
  parser.resetCache()
/usr/local/lib/python3.12/dist-packages/matplotlib/_mathtext.py:45: PyparsingDepr
ecationWarning: 'enablePackrat' deprecated - use 'enable_packrat'
  ParserElement.enablePackrat()
In /usr/local/lib/python3.12/dist-packages/matplotlib/mpl-data/stylelib/classic.m
plstyle: 'parseString' deprecated - use 'parse_string'
In /usr/local/lib/python3.12/dist-packages/matplotlib/mpl-data/stylelib/classic.m
plstyle: 'resetCache' deprecated - use 'reset_cache'
/usr/local/lib/python3.12/dist-packages/feast/ui_server.py:45: DeprecationWarnin
g:
        on_event is deprecated, use lifespan event handlers instead.

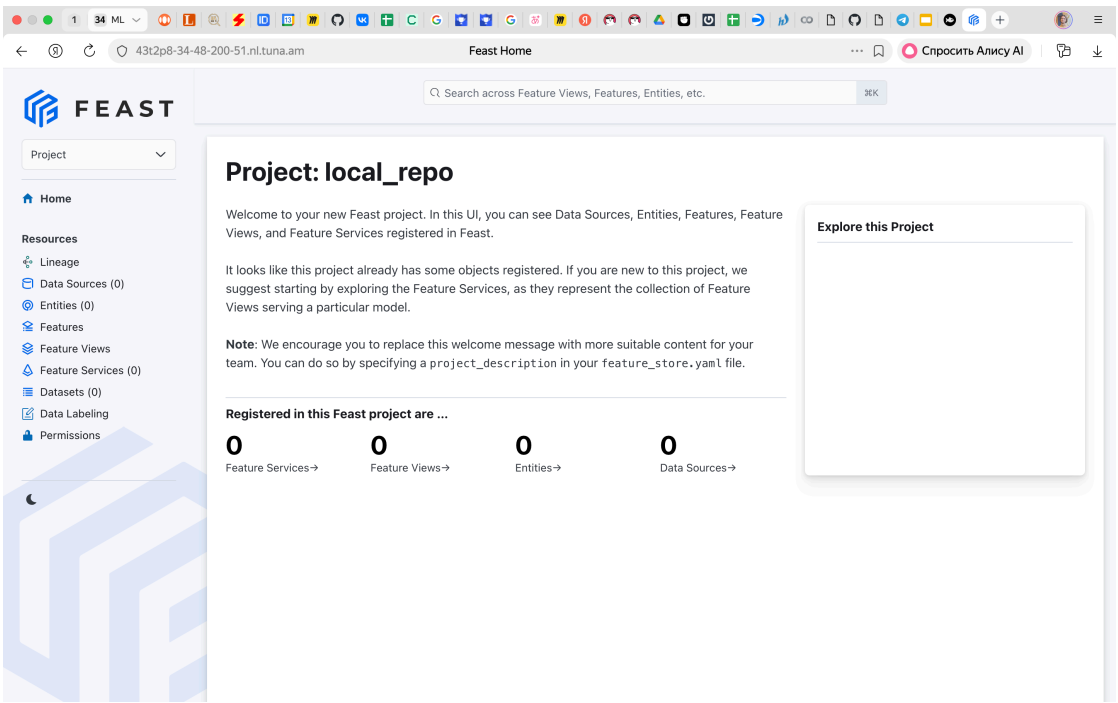
        Read more about it in the
        [FastAPI docs for Lifespan Events](https://fastapi.tiangolo.com/advanced/
events/).

  @app.on_event("shutdown")
/usr/local/lib/python3.12/dist-packages/fastapi/applications.py:4656: Deprecation
Warning:
        on_event is deprecated, use lifespan event handlers instead.

        Read more about it in the
        [FastAPI docs for Lifespan Events](https://fastapi.tiangolo.com/advanced/
events/).

  return self.router.on_event(event_type)  # ty: ignore[deprecated]
/usr/local/lib/python3.12/dist-packages/websockets/legacy/__init__.py:6: Deprecat
ionWarning: websockets.legacy is deprecated; see https://websockets.readthedocs.i
o/en/stable/howto/upgrade.html for upgrade instructions
  warnings.warn(  # deprecated in 14.0 - 2024-11-09
/usr/local/lib/python3.12/dist-packages/uvicorn/protocols/websockets/websockets_i
mpl.py:17: DeprecationWarning: websockets.server.WebSocketServerProtocol is depre
cated
  from websockets.server import WebSocketServerProtocol
INFO:     Started server process [63869]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
INFO:     Uvicorn running on http://0.0.0.0:8889 (Press CTRL+C to quit)
INFO:     151.247.224.92:0 - "GET / HTTP/1.1" 200 OK
INFO[14:29:54] GET / — 200 OK
INFO:     151.247.224.92:0 - "GET /static/css/main.d54c2a3e.css HTTP/1.1" 200 OK
INFO[14:29:54] GET /static/css/main.d54c2a3e.css — 200 OK
INFO:     151.247.224.92:0 - "GET /static/js/main.03ac5c90.js HTTP/1.1" 200 OK
INFO[14:29:55] GET /static/js/main.03ac5c90.js — 200 OK
INFO:     151.247.224.92:0 - "GET /static/media/feast-icon-grey.e5aa0a305d8efd775
c4b.svg HTTP/1.1" 200 OK
INFO[14:29:55] GET /static/media/feast-icon-grey.e5aa0a305d8efd775c4b.svg — 200 O
K
INFO:     151.247.224.92:0 - "GET /projects-list.json HTTP/1.1" 200 OK
INFO[14:30:11] GET /projects-list.json — 200 OK
INFO:     151.247.224.92:0 - "GET /static/js/icon.arrow_down.50a8ddfa.chunk.js HT
TP/1.1" 200 OK
INFO[14:30:11] GET /static/js/icon.arrow_down.50a8ddfa.chunk.js — 200 OK
```

```
INFO:     151.247.224.92:0 - "GET /registry HTTP/1.1" 200 OK
INFO[14:30:12] GET /registry - 200 OK
INFO:     151.247.224.92:0 - "GET /static/js/icon.app_graph.72cb7727.chunk.js HTT
P/1.1" 200 OK
INFO[14:30:12] GET /static/js/icon.app_graph.72cb7727.chunk.js - 200 OK
INFO:     151.247.224.92:0 - "GET /static/js/icon.documentEdit.8c4e6fd4.chunk.js
HTTP/1.1" 200 OK
INFO[14:30:12] GET /static/js/icon.documentEdit.8c4e6fd4.chunk.js - 200 OK
INFO:     151.247.224.92:0 - "GET /registry HTTP/1.1" 200 OK
INFO[14:30:12] GET /registry - 200 OK
INFO:     151.247.224.92:0 - "GET /static/js/icon.moon.94558c42.chunk.js HTTP/1.
1" 200 OK
INFO[14:30:12] GET /static/js/icon.moon.94558c42.chunk.js - 200 OK
INFO:     151.247.224.92:0 - "GET /static/js/icon.search.78d35d76.chunk.js HTTP/
1.1" 200 OK
INFO[14:30:13] GET /static/js/icon.search.78d35d76.chunk.js - 200 OK
INFO:     151.247.224.92:0 - "GET /feast-favicon-36x36.png HTTP/1.1" 200 OK
INFO[14:30:13] GET /feast-favicon-36x36.png - 200 OK
INFO:     Shutting down
INFO[14:31:37] Bye
INFO:     Waiting for application shutdown.
INFO:     Application shutdown complete.
INFO:     Finished server process [63869]
```



Мы создали базовую конфигурацию Feature Store.

В реальном проекте сюда добавляются:

- Data Sources - откуда брать данные (CSV, BigQuery, Parquet)
- Entities - ключевые сущности (user_id, product_id)
- Feature Views - наборы признаков
- Feature Services - группы признаков для модели

4. Обосновать готовность ML-системы к развертыванию в продуктивном окружении

В этом домашнем задании вы сохраняли разные версии данных, создали хранилище признаков для своей ML-системы.

Насколько ML-система готова к развёртыванию?

Ожидаемый артефакт: текст в текстовой ячейке

На основе выполненной работы система **частично готова** к продуктивному окружению.

Мы обеспечили воспроизводимость данных через **DVC** - пайплайн версионирован и может быть повторен в любой момент. Эксперименты отслеживаются в **MLflow**: все метрики, параметры и модели сохранены, к любой версии можно откатиться. Создано хранилище признаков на **Feast**, что позволяет централизованно использовать одни и те же признаки разными моделями.

Однако для полноценного production необходимо заменить локальное хранение на облачное (S3/MinIO для артефактов и PostgreSQL для MLflow), настроить CI/CD-пайплайн для автоматического запуска при изменениях, добавить мониторинг дрейфа данных и валидацию входящих признаков.

Ключевой риск - невоспроизводимость - уже устранен, система готова к staging-окружению.

5. Разработать схему ML-системы для размытия (заблюривания) лиц на изображениях

Вводные данные от бизнеса: нужно удалить биометрию клиентов с видео, и пример видеоданных.

КоАП РФ Статья 13.11.3 Непринятие организационных и технических мер по обеспечению безопасности биометрических персональных данных <...> - влечет наложение административного штрафа на должностных лиц в размере от трехсот тысяч до пятисот тысяч рублей; на юридических лиц - от одного миллиона до одного миллиона пятисот тысяч рублей.

Нужно ознакомиться с нижеприведенным кодом, осознать что он синхронный и однопоточный, подумать как можно использовать параллелизм данных, выбрать оптимальную архитектуру и составить схематическое описание ML-системы с помощью библиотеки Diagrams.

Ожидаемый артефакт: схематическое описание ML-системы ячейке

In [2…

```
import gdown; from IPython.display import Video, Image
gdrive_url = 'https://drive.google.com/uc?export=download&id=1OjSrdAS4zVfa1u7vtyh4
gdown.download(gdrive_url, quiet=True); Video("/content/HW5_Woman_Happy.mp4",embed
```

Out[2…

In [2…

```
gdrive_url = 'https://drive.google.com/uc?export=download&id=1UEEF9GIVaO2mRKxVFIal
gdown.download(gdrive_url, quiet=True); Image("/content/HW5_Woman_Happy_blurred.gi
```

Out[2…

```
<IPython.core.display.Image object>
```

%%html

Нижеприведенный код ML-системы обрабатывает каждый кадр видео так:

In [2…

```
!wget -q https://raw.githubusercontent.com/opencv/opencv/master/data/haarcascades/

import cv2
from google.colab.patches import cv2_imshow
import numpy as np

def apply_mosaic_effect(face_roi, pixel_size=10):
    if face_roi.shape[0] == 0 or face_roi.shape[1] == 0:
        return face_roi
    h, w, _ = face_roi.shape
    small_h = max(1, h // pixel_size)
    small_w = max(1, w // pixel_size)
    if small_h == 0: small_h = 1
    if small_w == 0: small_w = 1
    downscaled = cv2.resize(face_roi, (small_w, small_h), interpolation=cv2.INTER_
    mosaiced_face = cv2.resize(downscaled, (w, h), interpolation=cv2.INTER_NEAREST
    return mosaiced_face

output_video_path = '/content/HW5_Woman_Happy_blurred.mp4'
video_file_path = '/content/HW5_Woman_Happy.mp4'
video_capture = cv2.VideoCapture(video_file_path)
face_cascade = cv2.CascadeClassifier('/content/haarcascade_frontalface_default.xml

if not video_capture.isOpened():
    print("Error: Could not open video file.")
else:
    fps = int(video_capture.get(cv2.CAP_PROP_FPS))
    width = int(video_capture.get(cv2.CAP_PROP_FRAME_WIDTH))
    height = int(video_capture.get(cv2.CAP_PROP_FRAME_HEIGHT))
    frame_count = int(video_capture.get(cv2.CAP_PROP_FRAME_COUNT))
    fourcc = cv2.VideoWriter_fourcc(*'XVID')
    video_writer = cv2.VideoWriter(output_video_path, fourcc, fps, (width, height)

    print(f"Загружено видео с характеристиками: FPS={fps}, Width={width}, Height={
    print(f"VideoWriter начинает запись в файл: {output_video_path}")

    video_capture.set(cv2.CAP_PROP_POS_FRAMES, 0) # Reset video_capture to the beg
    with tqdm(total=frame_count, desc="обработано ") as pbar:
      while video_capture.isOpened():
          ret, frame = video_capture.read()
          if not ret:
              break
          pbar.update(1)

          if False: cv2_imshow(cv2.resize(frame, None, fx=0.05, fy=0.05, interpola

          gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
          faces = face_cascade.detectMultiScale(gray, 1.1, 4)

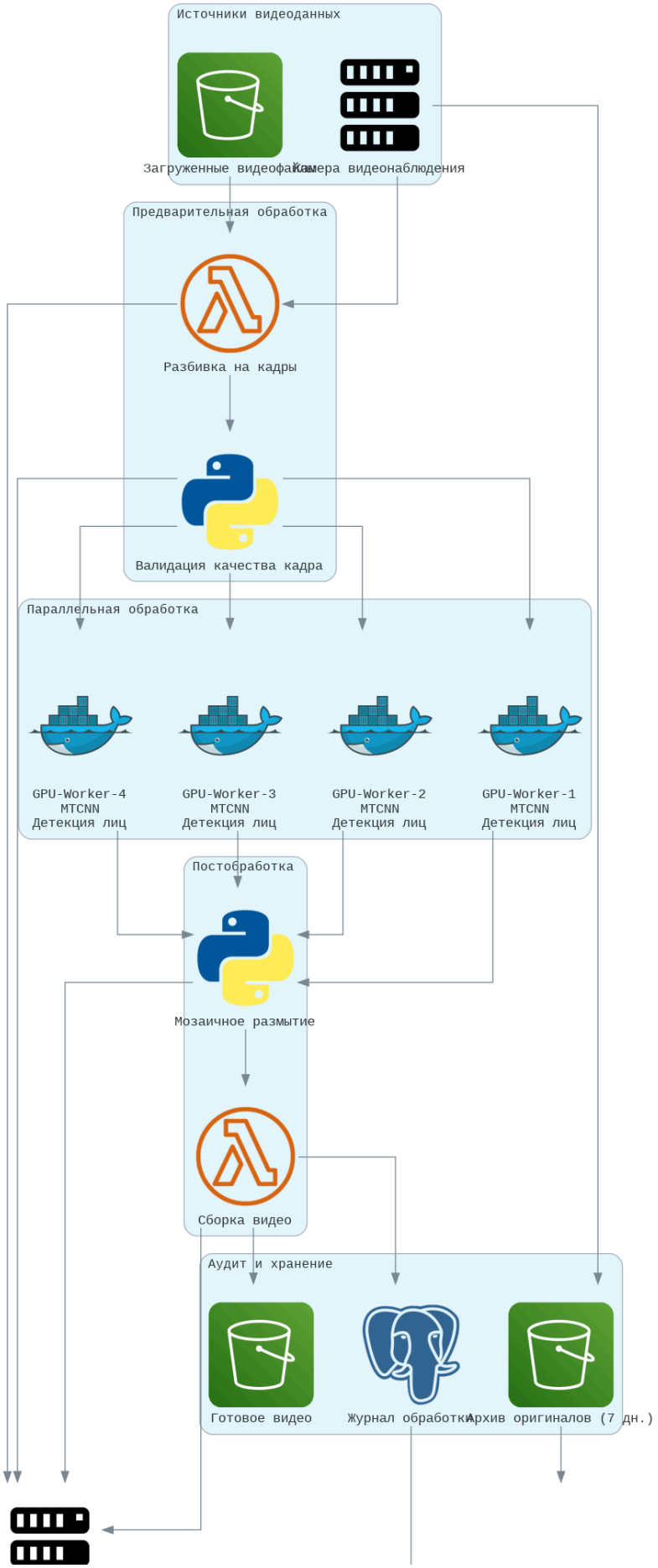
          for (x, y, w, h) in faces:
              face_roi = frame[y:y+h, x:x+w]
              mosaiced_face = apply_mosaic_effect(face_roi, pixel_size=100)
              frame[y:y+h, x:x+w] = mosaiced_face
          video_writer.write(frame)

    video_capture.release()
    video_writer.release()
    cv2.destroyAllWindows()

    print(f"\nVideoWriter записал видео в файл {output_video_path}")
```

```
Загружено видео с характеристиками: FPS=25, Width=3840, Height=2160
VideoWriter начинает запись в файл: /content/HW5_Woman_Happy_blurred.mp4
обработано : 100%|██████████| 204/204 [08:23<00:00,  2.47s/it]
VideoWriter записал видео в файл /content/HW5_Woman_Happy_blurred.mp4
```

Диаграмма ML-системы:


Источники видеоданных
Загруженные видеофайлы
Камера видеонаблюдения
Предварительная обработка
Разбивка на кадры
Валидация качества кадра
Параллельная обработка
GPU-Worker-4
MTCNN
Детекция лиц
GPU-Worker-3
MTCNN
Детекция лиц
GPU-Worker-2
MTCNN
Детекция лиц
GPU-Worker-1
MTCNN
Детекция лиц
Постобработка
Мозаичное размытие
Сборка видео
Аудит и хранение
Готовое видео
Журнал обработки
Архив оригиналов (7 дн.)

ML System for Face Blurring

In [2…

```
!pip install diagrams -q
from diagrams import Diagram, Edge, Cluster
from diagrams.onprem.queue import Kafka
from diagrams.onprem.compute import Server
from diagrams.programming.language import Python
from diagrams.aws.storage import S3
from diagrams.generic.storage import Storage
from diagrams.onprem.container import Docker
from diagrams.generic.blank import Blank
from diagrams.onprem.database import PostgreSQL
from diagrams.aws.compute import LambdaFunction

with Diagram("ML System for Face Blurring", show=False, direction="TB"):

    # Входные данные
    with Cluster("Источники видеоданных"):
        camera = Server("Камера видеонаблюдения")
        upload = S3("Загруженные видеофайлы")

    # Предобработка
    with Cluster("Предварительная обработка"):
        splitter = LambdaFunction("Разбивка на кадры")
        validator = Python("Валидация качества кадра")

    # Параллельная обработка
    with Cluster("Параллельная обработка"):
        gpu_workers = []
        for i in range(4):
            worker = Docker(f"GPU-Worker-{i+1}\nMTCNN\nДетекция лиц")
            gpu_workers.append(worker)

    # Постобработка
    with Cluster("Постобработка"):
        mosaic = Python("Мозаичное размытие")
        assembler = LambdaFunction("Сборка видео")

    # Аудит
    with Cluster("Аудит и хранение"):
        db = PostgreSQL("Журнал обработки")
        archive = S3("Архив оригиналов (7 дн.)")
        output = S3("Готовое видео")

    # Мониторинг
    monitor = Server("Prometheus: Метрики")

    # Потоки данных
    [camera, upload] >> splitter
    splitter >> validator
    validator >> gpu_workers[0]
    validator >> gpu_workers[1]
    validator >> gpu_workers[2]
    validator >> gpu_workers[3]

    for w in gpu_workers:
        w >> mosaic

    mosaic >> assembler
    assembler >> output
    assembler >> db

    camera >> archive
    archive >> Blank("Автоудаление через 7 дней")

    [splitter, validator, mosaic, assembler] >> monitor
    db >> monitor
```

```
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 34.5/34.5 MB 27.9 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 47.1/47.1 kB 2.4 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 226.5/226.5 kB 13.4 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 99.4/99.4 kB 6.2 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 5.8/5.8 MB 78.3 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 469.0/469.0 kB 28.0 MB/s eta 0:00:00
ERROR: pip's dependency resolver does not currently take into account all the pac
kages that are installed. This behaviour is the source of the following dependenc
y conflicts.
google-adk 1.29.0 requires tenacity<10.0.0,>=9.0.0, but you have tenacity 8.5.0 w
hich is incompatible.
```

6. Итоговое оформление

В итоговых выводах дайте 5–8 предложений о своем опыте работы с инструментами модуля: что оказалось простым, что вызвало трудности, какие выводы сделали о воспроизводимости эксперимента с DVC/MLflow в реальных проектах.

Работа с инструментами модуля оказалась неожиданно увлекательной.

Самым простым и интуитивно понятным стал **MLflow**: буквально несколько строк кода, и вот уже все метрики и модели аккуратно разложены.

DVC сначала показался сложнее из-за необходимости работать с Git и YAML-файлами, но когда пайплайн запустился и восстановил удаленные данные одной командой - пришло понимание, что такое воспроизводимость.

Feast оставил двойственное впечатление: создание YAML-конфига далось легко, а вот запуск UI через туннели Tuna превратился в неожиданный квест с лимитами и таймаутами, который, впрочем, многому научил.

Marimo стал самым приятным открытием - реактивные ноутбуки действительно решают проблему "грязного состояния", от которой страдают классические Jupyter-блокноты.

Главный вывод: воспроизводимость в ML - это необходимость, особенно когда работаешь в команде или возвращаешься к проекту спустя месяц.

Без DVC и MLflow любой эксперимент рискует остаться лишь строчкой в блокноте, которую невозможно повторить.

Теперь я понимаю, что внедрение этих инструментов в реальных проектах экономит не дни, а недели работы.